MIL-STD-1896(AT)
20 June 1985
Superseding
(see 6.1)

MILITARY STANDARD

JOINTS, BRAZED, INSPECTION OF

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

NO DELIVERABLE DATA DOCUMENT

AREA NDTI

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, DC 20301

Joints, Brazed, Inspection of.

MIL-STD-1896(AT)

1. This Military Standard is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

FOREWORD

This Military standard establishes the requirements for visual, penetrant, radiographic, and ultrasonic inspection. It specifies the procedures to be followed, records to be maintained, and the reports to be made.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

C\ ^TS

Paragraph			Page
	1.	SCOPE	1
	1.1	Purpose	1
	1.2	Scope	1
	2.	REFERENCED DOCUMENTS	2
	2.1	Issues of documents	2
	2.2	Other publications	2
	3.	DEFINITIONS	3
	3.1	Braze symbols	3
	3.2	Braze classification	3
	3.3	Void	3
	3.4	Lack of braze	3
	3.5	Lack of bond	3
	3.6	Surface porosity	3
	3.7	Wetted	3
	3.8	Other items	3
	4.	GENERAL REQUIREMENTS	4
	4.1	Acceptable criteria	4
	4.1.1	Pressure test	4
	5.	DETAIL REQUIREMENTS	5
	5.1	Test methods	5
	5.1.1	Visual	5
	5.1.2	Fluorescent penetrant	5
	5.1.3	Radiographic and ultrasonic	5
	5.1.4	Pressure test	5
	5.2	Rework	5
	5.2.1	Material removal	5
	5.2.2	Reinspection	6
	5.2.3	Records of rework	6
	6.	NOTES	7
	6.1	Supersession data	7

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Page

TABLES

Table I. Acceptance limits for visual and penetrant inspection of brazed joints..... 8

II. Acceptance limits for radiographic and ultrasonic inspection of brazed joints..... 9

FIGURES

Figure 1. Typical High Temperature Braze Joint..... 10

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:28Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Purpose. Inspection of brazed joints is performed to detect discontinuities which may lessen the integrity of the brazed parts.

1.2 Scope. This standard covers the requirements for non-destructive inspection of brazed joints used on aircraft quality parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL-I-6866	- Inspection, Penetrant, Method of.

STANDARDS
MILITARY

MIL-STD-453	- Inspection, Radiographic.
MIL-STD-1875	- Inspection, Ultrasonic, Requirements for.

(Copies of specifications, standards, drawings, and publications required by contractors in accordance with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	- Symbols for Welding and Nondestructive Testing, including Brazing.
AWS A3.0	- Welding Terms and Definitions (including Brazing).

(Copies of the above publications may be obtained from the American Welding Society, 2501 Northwest 7th Street, Miami, Florida 33125.)

(Technical society and technical association specifications and standards are generally available for reference from libraries, technical groups and using federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Braze symbols. The symbols used for identification of braze joints are in accordance with the Standard AWS A2.4.

3.2 Braze classification. The classification of brazed joints is as defined in the applicable process specification cited on the applicable engineering drawing for a specific part.

3.3 Void. A void is an interruption in the braze which is continuous through the joint cross section.

3.4 Lack of braze. Lack of braze is an interruption in the braze which is not continuous through the joint cross section.

3.5 Lack of bond. Lack of bond is an area of the braze joint where the braze alloy has failed to adhere (bond) to the base metal, which condition produces a signal response when inspected ultrasonically.

3.6 Surface porosity. Surface porostiy is a roughening or spongy appearance on the surface of the brazed alloy. Surface porosity is confined to the surface of the braze with no signigicant joint penetration as determined by radiographic inspection.

3.7 Wetted. A properly "wetted" braze joint is a joint where the braze alloy deposit is smooth, adherent, and with no repulsion of the braze alloy by the base metal.

3.8 Other terms. Terms for equipment, processes and procedures described in this standard can be found in the AWS A3.0 reference document.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:28Z
Check the source to verify that this is the current version before use.

4. GENERAL REQUIREMENTS

4.1 Acceptance criteria. Unless otherwise specified on the applicable engineering drawing or applicable inspection specifications for a specific part, the following criteria shall apply:

a. Cracks, voids, ring or fillet of braze material at all joint edges, and surface porosity regardless of location shall not be acceptable.

b. Internal defects shall be in accordance with the limits stated in table II.

c. An uninterrupted band of braze (excluding the fillet) shall be required around the circumference or along the length of braze joints.

d. The quality of braze workmanship shall be such that the surface of assembled details shall be free from excess braze material.

e. Braze joints not properly wetted (see 3.7) or without alloy are not acceptable. This is indicated by the following:

1) Blue or black coloration of base material in area adjacent to the joint after removal from braze furnace and prior to further processing (see fiugre 1).

2) Lumpy or uneven appearance of braze alloy.

3) Discoloration of braze alloy (i.e., not silver or other light even color typical of the braze alloy).

4.1.1 Pressure test. When required, pressure tests shall be in accordance with the requirements of the applicable engineering drawing for a specific part.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:28Z
Check the source to verify that this is the currentversion before use.

5. DETAIL REQUIREMENTS

5.1 Test methods. Inspection of brazed joints may be performed at the subassembly or assembly level provided the entire joint is accessible for inspection. Brazed parts requiring heat treatment shall be inspected after completion of all heat treat operations.

Note: Brazed joints inspected in-process prior to machining of the braze shall require visual and fluorescent penetrant inspection after machining operations.

5.1.1 Visual. All brazed joints shall be visually inspected to verify conformance to the requirements of 4.1.a, 4.1.c, and 4.1.d.

5.1.2 Fluorescent penetrant. All brazed joints shall be fluorescent penetrant inspected in accordance with Specification MIL-I-6866 to verify conformance to 4.1.a.

5.1.3 Radiographic and ultrasonic. Unless otherwise specified on the engineering drawing, all brazed joints shall be subject to radiographic or ultrasonic inspection (MIL-STD-453 or MIL-STD-1875) to verify conformance to 4.1.b and 4.1.c. The frequency of inspection shall be as follows:

a. Class I brazed joints. All class I brazed joints shall be radiographically or ultrasonically inspected.

b. Class II brazed joints. All class II brazed joints shall be radiographically or ultrasonically inspected until the quality level of the joint is established. If the brazing process is modified or changed, the radiographic or ultrasonic inspection shall be reinstated until the quality level of the joint is re-established.

c. Class III brazed joints. Class III brazed joints shall not require radiographic or ultrasonic inspection.

5.1.4 Pressure test. Pressure testing shall be performed to verify conformance to 4.1.1.

5.2 Rework. Braze rework on a part (assembly) may be performed upon approval of the acquisitioning authority. Braze rework procedures shall be approved by the acquisitioning authority. When a standard rework procedure does not exist, the part (assembly) shall be referred to the acquisitioning authority for disposition.

5.2.1 Material removal. Material removal (e.g., grinding) for braze rework shall be confined to the braze area. When the height of the braze bead causes interference with mating parts, the braze(s) may be ground to fit.

CAUTION: Care should be exercised to ensure that parent metal is not removed during metal removal operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

5.2.2 Reinspection. All braze rework shall be inspected in accordance with the applicable requirements of 5.1.

5.2.3 Records of rework. Records shall encompass all pertinent data relative to the reason for rework, location and size of defect, and the procedure used in rework.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Supersession data. This standard supersedes Avco Lycoming Division, Stratford, Connecticut, internal specification P6814C, 26 April 1982.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project No. NDTI-A029)

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

TABLE I. Acceptance limits for visual and penetrant inspection of brazed joints.

Type of defect	Class I	Class II	Class III
Cracks (braze and base mat'l)	U	U	U
Voids (ref 3.3)	U	U	U
Ring or fillet of braze mat'l at all joint edges	100% of C	90% of C	80% of C
Surface porosity (ref 3.6)	U	A	A

Notes:

U = Unacceptable

A = Acceptable

C = Circumference or length of brazed joint

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

TABLE II. Acceptance limits for radiographic and ultrasonic inspection of brazed joints.

Type of defect	Width 1/ (transverse to circumference or length of joint)	Max total (cumulative) length percentage of joint 2/	
		Class I	Class II
Cracks (braze and base mat'l)	----------	U	U
Lack of braze or bond (ref 3.4 and 3.5)	Greater than W/2	2 3/	3 3/
	W/2 to W/3	5	10
	W/3 to W/4	10	20
	W/4 to W/5	20	100
	Less than W/5	100	---
Voids (ref 3.3)	W/1	U	U

Notes:

1/. Definitions

W = Width of brazed joint
U = Unacceptable

2/. Maximum total (cumulative) length of joint defects:

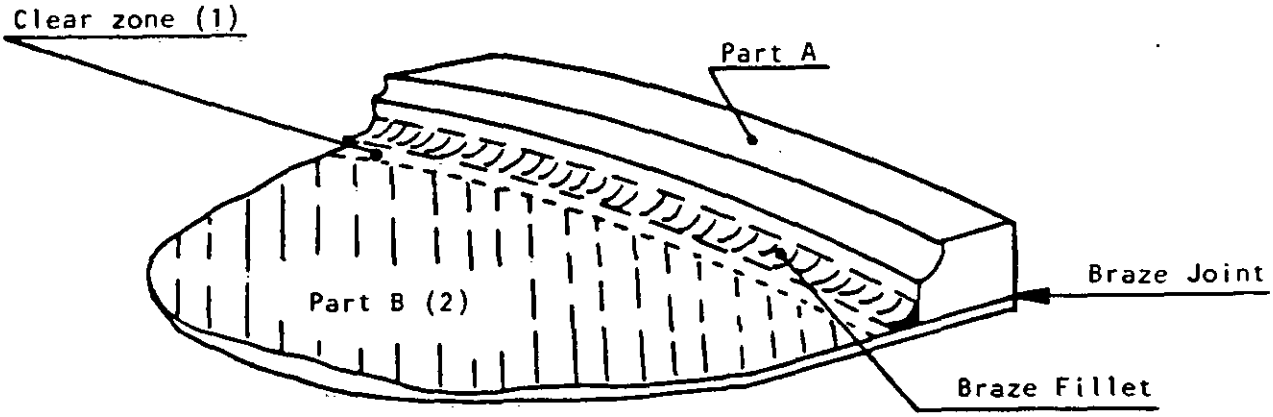
Example:

If 3 percent of Class I brazed joint indications fall into the W/2 to W/3 category, the allowable percentage for defects in the W/3 to W/4 category shall be 7 percent maximum.

3/. Lack of braze or bond greater than W/2 shall be accpetable when the following requirements can be met:

a. Individual indications greater than W/2 in width shall not exceed 1 percent in length.
b. For 0.5 inch (12.7 mm) on either side of the W/2 indication (per a.) there shall be no indications larger than W/4.
c. When two or more indications are closer to the W/2 indication than two times the width of the smaller indication, the indications and the space between them may be evaluated as a single W/2 indication (per a.) and accepted or rejected accordingly.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:28Z
Check the source to verify that this is the current version before use.



NOTES:

(1) A clear zone shown on part "B" immediately adjacent to the braze fillet denotes proper use of brazing flux and the satisfactory wetting of the base material by the braze alloy.

(2) The shading shown on part "B" represents the discoloration normally found on parts due to the effect of the braze furnace heat treat cycle.

FIGURE 1. Typical high temperature braze joint.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:28Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Tank-Automotive Command
ATTN: DRSTA-GSS
Warren, MI 48080

Source: https://assist.dla.mil -- Downloaded: 202640-24T01:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1896	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-24T01:28Z
Check the source to verify that this is the current version before use.